From Ising Model to Path Integral

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1 The Ising Model as a Prototype for Phase Transitions

1.1 Microscopic Definition

We consider a d-dimensional lattice with N sites. At each lattice site i, we assign a classical spin variable:

$$s_i \in \{+1, -1\}. \tag{1}$$

The Hamiltonian of the Ising model is given by

$$E[\{s_i\}] = -B \sum_i s_i - J \sum_{\langle ij \rangle} s_i s_j, \tag{2}$$

where:

- B is the external magnetic field,
- J is the interaction strength,
- $\langle ij \rangle$ denotes nearest-neighbor pairs.

We focus on the ferromagnetic case $J > 0$, where neighboring spins energetically prefer alignment.

Physical picture: The system is governed by a competition:

- Interaction (J): favors order,
- External field (B): biases direction,
- Temperature (T): favors disorder via entropy.

1.1.1 Partition Function and Thermodynamics

In the canonical ensemble:

$$Z(T, B) = \sum_{\{s_i\}} e^{-\beta E[\{s_i\}]}, \quad \beta = \frac{1}{T}. \tag{3}$$

The thermodynamic free energy is:

$$\langle F(T, B) \rangle = \langle E \rangle - TS = -T \ln Z. \tag{4}$$

As a reminder, we are familiar with this relationship of the canonical ensemble in statistical physics along with the followings:

1. $\langle E \rangle = -\frac{\partial}{\partial \beta} \ln Z$,
2. Given the probability distribution $p(n) = e^{-\beta E_n}/Z$
   $S = -\mathrm{Tr}(\hat{\rho} \ln \hat{\rho}) = -\sum_n p(n) \ln p(n) = -\frac{\partial}{\partial T}(T \ln Z)$,
3. $T \frac{\partial}{\partial T} = -\beta \frac{\partial}{\partial \beta}$.

Define the total magnetization:

$$M = \sum_i s_i, \tag{5}$$

and the **magnetization density** (order parameter):

$$m \equiv \frac{1}{N}\langle M \rangle \in [-1, 1]. \tag{6}$$

Using standard thermodynamic identities:

$$m = \frac{1}{N\beta}\frac{\partial \ln Z}{\partial B}. \tag{7}$$

Key point: The partition function Z encodes all thermodynamic information. Phase transitions arise from its non-analytic behavior in the thermodynamic limit $N \to \infty$.

1.2 Reformulation: Effective Free Energy $F(m)$

We reorganize the partition function by grouping configurations with fixed magnetization density m:

$$Z = \sum_m \sum_{\{s_i\}|m} e^{-\beta E[\{s_i\}]} \equiv \sum_m e^{-\beta F(m)}. \tag{8}$$

Here $F(m)$ is an **effective free energy** as a function of the coarse variable m. There some subtleties in the derivation of the above relation that need further illustration. We can rewrite:

$$\sum_{\{s_i\}|m} e^{-\beta E[\{s_i\}]} = \Omega(m) \left\langle e^{-\beta E} \right\rangle_m, \tag{9}$$

where $\Omega(m)$ is number of configurations with a fixed magnetization m and the corresponding entropy $S(m) = \ln \Omega(m)$. Thus it yields the same relation as the thermodynamic free energy:

$$\begin{aligned} &-\beta F(m) = \ln \Omega(m) - \beta E(m) \\ &\Rightarrow F(m) = E(m) - T \ln \Omega(m) = E(m) - TS(m). \end{aligned} \tag{10}$$

This means that the effective free energy can be interpreted as the competition between energy and entropy at fixed order parameter. Rewriting the partition function in terms of $F(m)$ is actually a definition and reinterpretation.

In the large N limit, we define the free energy density:

$$f(m) = \frac{F(m)}{N}. \tag{11}$$

Then:

$$Z \approx \int dm\ e^{-\beta N f(m)}. \tag{12}$$

For large N, the integral is dominated by the saddle point:

$$\left.\frac{df}{dm}\right|_{m=m_{\min}} = 0. \tag{13}$$

Thus the value of the effective free energy $F(m)$ taken at the saddle point yields the thermodynamic free energy:

$$\langle F \rangle \approx N f(m_{\min}). \tag{14}$$

Interpretation: This is the first step toward a field-theoretic description: we replace microscopic degrees of freedom $\{s_i\}$ with a macroscopic variable m.

1.3 Mean Field Approximation

We approximate interactions by replacing each spin with its average:

$$\langle s_i \rangle \approx m. \tag{15}$$

Then:

$$\frac{E}{N} \approx -Bm - \frac{1}{2}Jqm^2, \tag{16}$$

where q is the number of nearest neighbors.

Next, we count the number of configurations with magnetization m. Let:

$$N_\uparrow = \frac{N(1+m)}{2}, \quad N_\downarrow = \frac{N(1-m)}{2}. \tag{17}$$

The number of configurations (multiplicity) is:

$$\Omega(m) = \frac{N!}{N_\uparrow! N_\downarrow!} = \frac{N!}{N_\uparrow!(N-N_\uparrow)!}. \tag{18}$$

Using Stirling's approximation ($\ln n! = n \ln n - n + O(\ln n)$):

$$\frac{1}{N} \ln \Omega(m) = \ln 2 - \frac{1}{2}(1+m)\ln(1+m) - \frac{1}{2}(1-m)\ln(1-m). \tag{19}$$

Thus the free energy density becomes:

$$f(m) = -Bm - \frac{1}{2}Jqm^2 - Ts(m), \tag{20}$$

where the entropy density ($S = \ln \Omega$) is:

$$s(m) = \ln 2 - \frac{1}{2}(1+m)\ln(1+m) - \frac{1}{2}(1-m)\ln(1-m). \tag{21}$$

1.3.1 Self-Consistency Equation

Minimizing $f(m)$:

$$\frac{df}{dm} = 0 \tag{22}$$

yields:

$$\begin{aligned} \beta(B + Jqm) &= \frac{1}{2}\ln\frac{1+m}{1-m} \\ \Rightarrow m &= \tanh(\beta B + \beta Jqm). \end{aligned} \tag{23}$$

Physical interpretation: Each spin experiences an *effective field*

$$B_{\text{eff}} = B + Jqm, \tag{24}$$

which includes contributions from neighboring spins. This is the essence of mean field theory.

1.3.2 From Mean Field Theory to the Critical Point

We start from the mean field self-consistency equation:

$$m = \tanh(\beta B + \beta Jqm). \tag{25}$$

To understand phase structure, we first set $B = 0$:

$$m = \tanh(\beta Jqm). \tag{26}$$

We analyze this equation near $m = 0$. Expanding the right-hand side:

$$\tanh x = x - \frac{x^3}{3} + \mathcal{O}(x^5). \tag{27}$$

Thus:

$$m = \beta Jqm - \frac{1}{3}(\beta Jqm)^3 + \cdots . \tag{28}$$

Rewriting:

$$m\,(1 - \beta Jq) + \frac{1}{3}(\beta Jq)^3 m^3 = 0 \tag{29}$$

A non-trivial solution $m \neq 0$ exists only when:

$$\beta Jq > 1. \tag{30}$$

This defines the **critical temperature**:

$$T_c = Jq. \tag{31}$$

Physical meaning:

- $T > T_c$: only solution is $m = 0$ (disordered phase)
- $T < T_c$: new solutions $m \neq 0$ emerge (ordered phase)

This marks a continuous (second-order) phase transition.

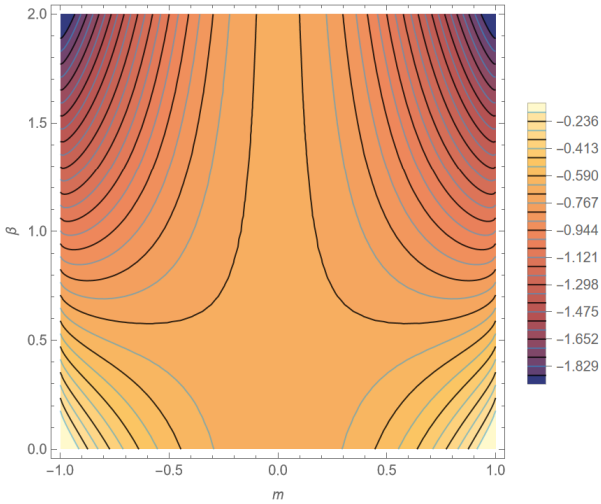

Figure 1: Contour plot of the free energy density, taking $J = 1,\ q = 2$. It is clear to see the behavior of the free energy changes at the critical point.

1.4 Landau Expansion from Mean Field Theory

Instead of solving the transcendental equation directly, we construct the free energy $f(m)$ and analyze its minima.

From mean field theory:

$$f(m) = -Bm - \frac{1}{2}Jqm^2 - Ts(m), \tag{32}$$

where:

$$s(m) = \log 2 - \frac{1}{2}(1+m)\log(1+m) - \frac{1}{2}(1-m)\log(1-m). \tag{33}$$

We expand $s(m)$ around $m = 0$:

$$s(m) = \log 2 - \frac{1}{2}m^2 - \frac{1}{12}m^4 + \mathcal{O}(m^6). \tag{34}$$

Substituting into $f(m)$:

$$f(m) = -Bm + \frac{1}{2}(T - Jq)m^2 + \frac{1}{12}Tm^4 + \cdots . \tag{35}$$

This is the **Landau free energy expansion**:

$$f(m) = -Bm + \frac{1}{2}a(T)m^2 + \frac{1}{4}b(T)m^4 + \cdots, \tag{36}$$

with:

$$a(T) = T - T_c, \quad b(T) = \frac{T}{3} > 0. \tag{37}$$

Key insight: The phase transition is controlled entirely by the sign change of $a(T)$.

1.4.1 Minimization and Phase Structure

We determine equilibrium magnetization by minimizing:

$$\frac{df}{dm} = 0. \tag{38}$$

This gives:

$$-B + a(T)m + b(T)m^3 = 0. \tag{39}$$

For simplicity, we only discuss the case of $B = 0$ here. If $B \neq 0$, the phase transition will not happen in $d = 1$ dimensions. In $B = 0$ case:

$$m\left[a(T) + b(T)m^2\right] = 0, \tag{40}$$

the solutions are:

$$m = 0, \quad m = \pm\sqrt{-\frac{a(T)}{b(T)}}. \tag{41}$$

Thus:

$$m = \begin{cases} 0 & T > T_c \\ \pm\sqrt{\frac{3(T_c - T)}{T}} & T < T_c \end{cases} \tag{42}$$

This shows that m grows continuously from zero, characteristic of a second-order phase transition.

1.4.2 Critical Exponents: Explicit Derivations

(1) Order Parameter Exponent β

Near T_c:

$$m \sim (T_c - T)^{1/2}. \tag{43}$$

Easy to get:

$$\beta = \frac{1}{2}. \tag{44}$$

(2) Susceptibility χ and Exponent γ

Define:

$$\chi = \left.\frac{\partial m}{\partial B}\right|_{B=0}. \tag{45}$$

From:

$$-B + a(T)m + b(T)m^3 = 0. \tag{46}$$

For $T > T_c$, m is small, so:

$$m \approx \frac{B}{a(T)} = \frac{B}{T - T_c}. \tag{47}$$

Thus:

$$\chi \sim \frac{1}{T - T_c}. \tag{48}$$

For $T < T_c$, expand around m_0:

$$m = m_0 + \delta m. \tag{49}$$

Linearizing:

$$\chi \sim \frac{1}{|T - T_c|}. \tag{50}$$

Therefore:

$$\gamma = 1. \tag{51}$$

(3) Critical Isotherm and Exponent δ

At $T = T_c$, we have:

$$-B + b(T_c)m^3 = 0. \tag{52}$$

Thus:

$$m \sim B^{1/3}. \tag{53}$$

So:

$$\delta = 3. \tag{54}$$

(4) Heat Capacity and Exponent α

The free energy at equilibrium:

$$F \approx Nf(m_{\min}). \tag{55}$$

For $T > T_c$:

$$m = 0 \quad \Rightarrow \quad f = 0. \tag{56}$$

For $T < T_c$:

$$f(m_0) \sim -\frac{(T_c - T)^2}{T}. \tag{57}$$

Thus:

$$C = -T\frac{\partial^2 F}{\partial T^2}. \tag{58}$$

gives a finite jump:

$$C \sim \text{const}. \tag{59}$$

Hence:

$$\alpha = 0 \tag{60}$$

1.5 Universality and the Need for a Field Description

The critical exponents derived above do not depend on microscopic details such as lattice structure or interaction strength. This leads to the concept of **universality**:

- Systems with different microscopic physics exhibit identical critical behavior
- Only symmetry and dimensionality determine the universality class

Conceptual implication:

Near the critical point:

- Correlation length diverges
- Fluctuations occur on all scales
- Microscopic details become irrelevant

This invalidates mean field theory and motivates a new description.

Key transition:

Instead of a single global order parameter m, we introduce a spatially varying field:

$$m \to m(x) \tag{61}$$

The partition function is then promoted to a functional integral:

$$Z = \int \mathcal{D}m(x)\, e^{-\beta F[m(x)]} \tag{62}$$

This is the starting point of **Landau-Ginzburg theory**, which provides the bridge from statistical mechanics to field theory and renormalization group.

2 Landau-Ginzburg Theory and the Emergence of a Field Description

2.1 Breakdown of the Mean Field Description

In the Landau mean field approximation, the system is described by a single global order parameter:

$$m = \frac{1}{N} \sum_i s_i. \tag{63}$$

This implicitly assumes that the system is spatially uniform. However, this assumption breaks down in the vicinity of the critical point. To illustrate this, we introduce the connected two-point correlation function:

$$G(r) = \langle s(0)s(r) \rangle - \langle s \rangle^2. \tag{64}$$

Although this will be discussed in detail later, we consider the correlation function here in terms of its physical significance. Empirically and theoretically, it behaves as follows:

$$G(r) \sim e^{-r/\xi}, \tag{65}$$

where ξ is the **correlation length**. As the temperature approaches the critical point:

$$\xi \to \infty \quad \text{as} \quad T \to T_c. \tag{66}$$

Physical implication:

- Fluctuations occur on arbitrarily large length scales
- The system can no longer be described by a single global variable
- Spatial variations of the order parameter become essential

This motivates the introduction of a **coarse-grained field**:

$$m \to m(x). \tag{67}$$

2.2 Coarse-Graining and the Emergence of a Functional Integral

We partition the lattice into blocks of size ℓ and define

$$m(x) = \frac{1}{\ell^d} \sum_{i \in \text{block}(x)} s_i, \tag{68}$$

with ℓ chosen sufficiently large that $m(x)$ can assume any value in $[-1, 1]$, yet sufficiently small that x may be treated as a continuous variable. This coarse-graining of m yields a smooth, spatially varying field.

The partition function can now be rewritten by inserting a constraint that enforces this coarse-grained definition:

$$Z = \int \mathcal{D}m(x) \sum_{\{s_i\}} \delta\big(m(x) - m_{\text{coarse}}[s]\big)\, e^{-\beta E[s]}. \tag{69}$$

We define an effective free energy functional:

$$e^{-\beta F[m(x)]} \equiv \sum_{\{s_i\}} \delta(\cdots)\, e^{-\beta E[s]}. \tag{70}$$

Thus, the partition function becomes:

$$Z = \int \mathcal{D}m(x)\, e^{-\beta F[m(x)]}. \tag{71}$$

Interpretation: This is a functional integral over field configurations. It is the statistical mechanics analogue of the Euclidean path integral in quantum field theory.

2.3 Constructing the Landau-Ginzburg Functional

We now determine the general form of $F[m(x)]$ using symmetry and locality.

1. Symmetry Constraints

The Ising model has a $\mathbb{Z}_2$ symmetry:

$$m(x) \to -m(x). \tag{72}$$

Therefore, the free energy must be invariant under this transformation. This forbids odd powers of m when $B = 0$.

2. Locality and Gradient Expansion

We assume that the free energy is local:

$$F[m] = \int d^d x \, \mathcal{F}[m(x), \nabla m(x), \cdots]. \tag{73}$$

Near the critical point, the field varies slowly, so we expand in gradients:

$$\mathcal{F} = \frac{1}{2}(\nabla m)^2 + V(m). \tag{74}$$

where:

$$V(m) = \frac{1}{2}a(T)m^2 + \frac{1}{4}b(T)m^4 + \cdots . \tag{75}$$

Thus, the Landau-Ginzburg free energy functional is

$$F[m] = \int d^d x \left[\frac{1}{2}(\nabla m)^2 + \frac{1}{2}a(T)m^2 + \frac{1}{4}b(T)m^4 - Bm\right], \tag{76}$$

where the case $B \neq 0$ has also been included. In the Ising model, the parameters are given by $a = T - T_c,\ b = T/3$ and $T_c = Jq$.

2.3.1 Origin of the Gradient Term

The term $(\nabla m)^2$ encodes the energetic cost of spatial inhomogeneity. To understand its origin, consider the lattice interaction:

$$J \sum_{\langle ij \rangle} s_i s_j. \tag{77}$$

In the coarse-grained description:

$$s_i \to m(x). \tag{78}$$

Expanding:

$$m(x+a) = m(x) + a\nabla m + \frac{a^2}{2}\nabla^2 m + \cdots . \tag{79}$$

Then:

$$m(x)m(x+a) = m^2(x) + am\nabla m + \frac{a^2}{2}m\nabla^2 m + \cdots . \tag{80}$$

After summing over space:

- Linear gradient terms vanish by symmetry
- Second-order terms generate $(\nabla m)^2$

Physical interpretation: Rapid spatial variations of the order parameter increase the free energy. The system prefers smooth configurations.

2.4 Uniform Saddle Point and Recovery of Mean Field Theory

Before studying fluctuations, we first verify that Landau-Ginzburg theory reproduces the mean field results at $B = 0$. The saddle point approximation provides a suitable starting point. The variational method helps identify the minima of the functional $F[m(x)]$:

$$\begin{aligned}
\delta F[m] &= \int d^d x \left[\nabla m \cdot \nabla \delta m + a(T) m \delta m + b(T) m^3 \delta m\right] \\
&= \int d^d x \left[-\nabla^2 m + a(T) m + b(T) m^3\right] \delta m \\
\Rightarrow \frac{\delta F[m]}{\delta m(x)} &= -\nabla^2 m + a(T) m + b(T) m^3.
\end{aligned} \tag{81}$$

The minimum of the free energy yields the Euler–Lagrange equation:

$$\nabla^2 m = a(T) m + b(T) m^3. \tag{82}$$

Clearly, spatially uniform configurations are the simplest solutions:

$$m(x) = m = \text{constant}. \tag{83}$$

We obtain:

$$a(T) m + b(T) m^3 = 0, \tag{84}$$

which is identical to the Landau equation obtained previously. Therefore:

$$m = \begin{cases} 0, & T > T_c, \\ \pm\sqrt{-\frac{a(T)}{b(T)}}, & T < T_c. \end{cases} \tag{85}$$

Conclusion: Landau-Ginzburg theory reproduces mean field results at the level of uniform configurations.

2.5 Beyond the Mean Filed Approximation

Unlike mean field theory, Landau-Ginzburg theory allows for spatially varying solutions. For $T < T_c$, the potential:

$$V(m) = \frac{1}{2} a(T) m^2 + \frac{1}{4} b m^4, \tag{86}$$

has two degenerate minima:

$$m = \pm m_0, \quad m_0 = \sqrt{-\frac{a(T)}{b(T)}}. \tag{87}$$

Different spatial regions may choose different minima, forming domains.

To study the interface between domains, we consider a one-dimensional configuration $m(x)$ minimizing:

$$F[m] = \int dx \left[\frac{1}{2}\left(\frac{dm}{dx}\right)^2 + V(m)\right]. \tag{88}$$

The Euler-Lagrange equation is:

$$\frac{d^2 m}{dx^2} = \frac{dV}{dm}. \tag{89}$$

Substituting $V(m)$:

$$\frac{d^2 m}{dx^2} = a(T) m + b(T) m^3. \tag{90}$$

Multiplying both sides by $\frac{dm}{dx}$:

$$\frac{d}{dx}\left[\frac{1}{2}\left(\frac{dm}{dx}\right)^2 - \frac{1}{2} a(T) m^2 - \frac{1}{4} b m^4\right] = 0. \tag{91}$$

Integrating:

$$\frac{1}{2}\left(\frac{dm}{dx}\right)^2 = V(m) - C. \tag{92}$$

For a single domain wall connecting $m = -m_0$ at $x \to -\infty$ and $m = +m_0$ at $x \to +\infty$, the gradient $\frac{dm}{dx}$ must vanish at infinity. Thus, C is chosen as $C = V(m_0)$ such that the right side is zero at $m = m_0$. Substituting this back, we get:

$$\left(\frac{dm}{dx}\right)^2 = \frac{b}{2}\left(\frac{-a}{b} - m^2\right)^2 = \frac{b}{2}\left(m^2 - m_0^2\right)^2. \tag{93}$$

Taking the square root and integrating:

$$\int \frac{dm}{m_0^2 - m^2} = \sqrt{\frac{b}{2}} \int dx. \tag{94}$$

Solving this equation yields the domain wall profile:

$$m(x) = m_0 \tanh\left(\frac{x}{\sqrt{2}\,\xi}\right), \tag{95}$$

where:

$$\xi = \frac{1}{\sqrt{|a(T)|}} = \frac{1}{\sqrt{T_c - T}}, \tag{96}$$

is the wall thickness. Near T_c, $\xi \to \infty$, so interfaces become diffuse.

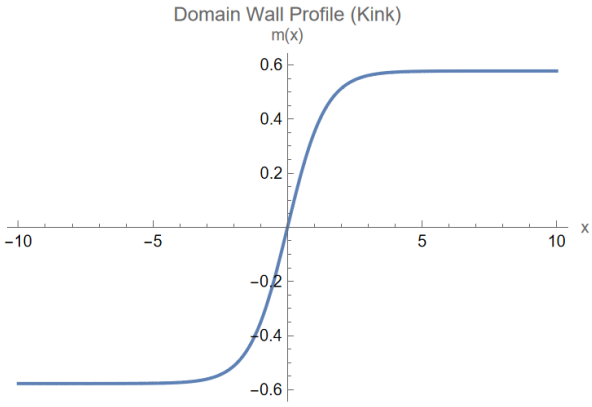

Figure 2: The domain wall in the $d = 1$ dimensions case, which interpolates smoothly between $-m_0$ and $+m_0$. Its thickness is set by ξ,

New physics beyond mean field:

- Existence of spatial structures (domains),
- Finite-energy topological configurations,
- Emergence of a characteristic length scale ξ.

2.6 Generating Functional and Correlation Functions

To compute correlation functions systematically, it is convenient to introduce a source term, following the standard formalism of quantum field theory.

We define the generating functional:

$$Z[J] = \int \mathcal{D}m \, \exp\left(-\tilde{S}[m] + \int d^d x \, J(x) m(x)\right). \tag{97}$$

Here:

- $\tilde{S}[m] = \beta F[m]$ plays the role of the Euclidean action,
- $J(x)$ is an external source coupled linearly to the field.

The original partition function is recovered as:

$$Z = Z[J]\big|_{J=0}. \tag{98}$$

Correlation functions are obtained by functional differentiation:

$$\langle m(x_1)\cdots m(x_n)\rangle = \frac{1}{Z[0]}\,\frac{\delta^n Z[J]}{\delta J(x_1)\cdots\delta J(x_n)}\bigg|_{J=0}. \tag{99}$$

In particular, the two-point function is:

$$G(x-y) = \langle m(x)m(y)\rangle = \frac{1}{Z[0]}\,\frac{\delta^2 Z[J]}{\delta J(x)\delta J(y)}\bigg|_{J=0}. \tag{100}$$

2.6.1 Evaluation of the Gaussian Functional Integral

We now compute $Z[J]$ explicitly. We first expand the field in Fourier modes:

$$m(x) = \int \frac{d^dk}{(2\pi)^d}\, m(k)e^{ik\cdot x}. \tag{101}$$

Since $m(x)$ is a real function, we have $m(-k) = m(k)^*$. Then:

$$\int d^dx\,(\nabla m)^2 = \int \frac{d^dk}{(2\pi)^d}\, k^2|m(k)|^2, \tag{102}$$

and:

$$\int d^dx\, m^2 = \int \frac{d^dk}{(2\pi)^d}\, |m(k)|^2. \tag{103}$$

Thus, in momentum space, the action is:

$$\tilde{S}[m] = \frac{\beta}{2}\int \frac{d^dk}{(2\pi)^d}\,\big(k^2 + a(T)\big)\,|m(k)|^2. \tag{104}$$

It is also possible to express its discrete form:

$$\tilde{S}[m] = \frac{\beta}{2V}\sum_{k,k'} m_k\left[k^2 + a(T)\right] m_{k'}\delta_{k,-k'}. \tag{105}$$

The source term becomes:

$$\int d^dx\, J(x)m(x) = \int \frac{d^dk}{(2\pi)^d} J(-k)m(k). \tag{106}$$

Thus:

$$Z[J] = \int \mathcal{D}m\ \exp\left[-\frac{\beta}{2}\int \frac{d^dk}{(2\pi)^d}(k^2+a)m(k)m(-k) + \int \frac{d^dk}{(2\pi)^d} J(-k)m(k)\right]. \tag{107}$$

This is a Gaussian functional integral. Completing the square:

$$m(k) = \tilde{m}(k) + \frac{1}{\beta}\frac{J(k)}{k^2 + a(T)}, \tag{108}$$

we obtain:

$$Z[J] = Z[0]\exp\left[\frac{1}{2\beta}\int \frac{d^dk}{(2\pi)^d}\frac{J(k)J(-k)}{k^2+a(T)}\right]. \tag{109}$$

The generating functional is quadratic in J, reflecting the **Gaussian nature** of the theory.

Taking functional derivatives:

$$\langle m(k)m(k')\rangle = \frac{1}{Z[0]}\,\frac{\delta^2 Z[J]}{\delta J(k)\delta J(k')}\bigg|_{J=0} = \frac{1}{\beta}(2\pi)^d\delta^{(d)}(k+k')\frac{1}{k^2+a(T)}. \tag{110}$$

This is the **Green's function** associated with the operator $-\nabla^2 + a(T)$. It is directly analogous to the propagator in QFT. Returning to position space, one readily verifies that the correlation function satisfies:

$$\left[-\nabla^2 + a(T)\right] G(x-y) = \frac{1}{\beta}\delta^{(d)}(x-y). \tag{111}$$

Key insight: The correlation function is the inverse of the quadratic operator in the action.

2.6.2 Position-Space Correlation Function and Correlation Length

We now compute:

$$G(r) = \langle m(r)m(0)\rangle = \frac{1}{\beta}\int \frac{d^dk}{(2\pi)^d}\frac{e^{ik\cdot r}}{k^2+a(T)}. \tag{112}$$

This integral can be evaluated using standard methods.

We exploit rotational symmetry and write:

$$G(r) = \frac{1}{\beta}\int \frac{d^dk}{(2\pi)^d}\frac{e^{ikr\cos\theta}}{k^2+a(T)}. \tag{113}$$

Performing the angular integration:

$$G(r) \propto \frac{1}{r^{(d-2)/2}}\int_0^\infty dk\,\frac{k^{d/2}}{k^2+a(T)}J_{(d-2)/2}(kr), \tag{114}$$

where J_ν is a Bessel function. For large r, the integral is dominated by the pole at:

$$k^2 = -a(T). \tag{115}$$

Thus:

$$G(r) \sim \frac{e^{-r/\xi}}{r^{(d-1)/2}}. \tag{116}$$

2.6.3 Correlation Length

We identify:

$$\xi = \frac{1}{\sqrt{a(T)}}. \tag{117}$$

Near the critical point:

$$a(T) \sim T - T_c, \tag{118}$$

so:

$$\xi \sim |T - T_c|^{-1/2}. \tag{119}$$

Physical Interpretation

The correlation function exhibits two key features:

- Exponential decay with characteristic length ξ,
- Power-law prefactor reflecting dimensionality.

As $T \to T_c$:

$$\xi \to \infty, \tag{120}$$

which implies:

- Long-range correlations,
- Absence of a characteristic scale,
- Emergence of scale invariance.

Finally, we briefly discuss the connection to field theory. It is evident that $G(k) = \left[k^2 + a(T)\right]^{-1}$ is the propagator with which we are familiar. Analogous to the massive scalar propagator, ξ^{-2} plays the role of the mass squared in this theory. As $\xi^{-1} \to 0$ when approaching the critical point, the critical point corresponds to a massless theory.

3 First-Principles MC Simulation of the 1D Ising Model

The goal of first-principles simulation is to compute statistical observables directly from the microscopic definition of the model, without introducing additional approximations.

The Ising Hamiltonian is given by:

$$E[s_i] = -J \sum_{\langle ij \rangle} s_i s_j - B \sum_i s_i. \tag{121}$$

Thermal expectation values are defined as:

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \sum_{s_i} \mathcal{O}[s_i] e^{-\beta E[s_i]}. \tag{122}$$

Since the number of configurations grows as 2^N, direct summation is intractable. Instead, we use Monte Carlo sampling.

3.1 Metropolis Algorithm and Markov Chain Sampling

The Metropolis algorithm constructs a Markov chain whose stationary distribution is the Boltzmann distribution:

$$P[s_i] = \frac{1}{Z} e^{-\beta E[s_i]}. \tag{123}$$

A single update consists of:

1. Randomly selecting a site i,
2. Proposing a spin flip $s_i \to -s_i$,
3. Computing the energy change:

$$\Delta E = 2J s_i (s_{i-1} + s_{i+1}) + 2B s_i, \tag{124}$$

4. Accepting the move with probability:

$$P_{\text{accept}} = \min(1, e^{-\beta \Delta E}). \tag{125}$$

This satisfies detailed balance and ensures convergence to the correct equilibrium distribution.

3.2 Time Averages and Observables

The system does not converge to a single configuration but instead samples the equilibrium distribution. Observables are computed as time averages:

$$\langle \mathcal{O} \rangle \approx \frac{1}{M} \sum_{t=1}^{M} \mathcal{O}(t). \tag{126}$$

Key observables include:

- Magnetization:

$$m = \frac{1}{N} \sum_i s_i, \tag{127}$$

- Energy per spin:

$$e = \frac{E}{N}, \tag{128}$$

- Correlation function:

$$G(r) = \langle s_i s_{i+r} \rangle. \tag{129}$$

3.3 Absence of Phase Transition in One Dimension

In one dimension with $B = 0$, the Ising model does not exhibit a true phase transition at finite temperature.

This can be understood from the competition between energy and entropy of domain walls. The free energy cost behaves as:

$$F \sim 2J - T \log N, \tag{130}$$

which becomes negative for any $T > 0$ in the thermodynamic limit. As a result, long-range order is destroyed.

However, the correlation length:

$$\xi \sim e^{2\beta J}, \tag{131}$$

can become very large at low temperatures, leading to apparent ordering over finite length scales.

3.4 Structure of the Correlation Function

The correlation function admits the decomposition:

$$G(r) = \langle s_i \rangle^2 + \langle \delta s_i \delta s_{i+r} \rangle. \tag{132}$$

- For $B \neq 0$, $\langle s_i \rangle \neq 0$, so:

$$G(r) \to m^2 \quad (r \to \infty), \tag{133}$$

  leading to a plateau.

- For $B = 0$, $\langle s_i \rangle = 0$, and:

$$G(r) \sim e^{-r/\xi}. \tag{134}$$

Notice that finite-size effects and statistical noise may produce an apparent flattening of $G(r)$ at large distances. In our simulation, a chain comprising 1000 sites is considered.

3.5 Numerical Results

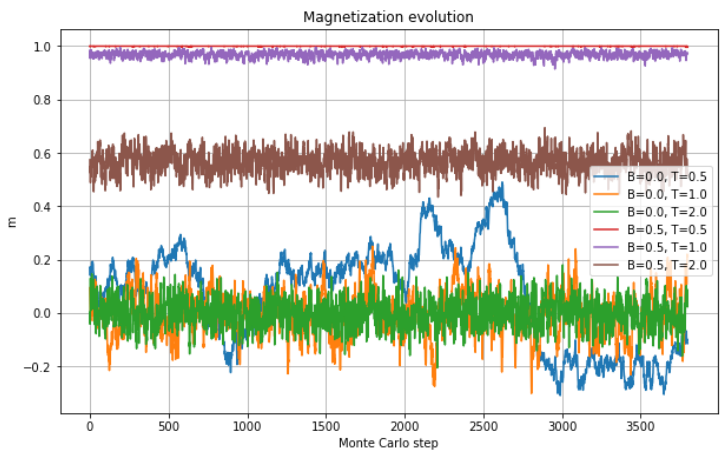

Figure 3: Magnetization evolution for different temperatures and magnetic fields.

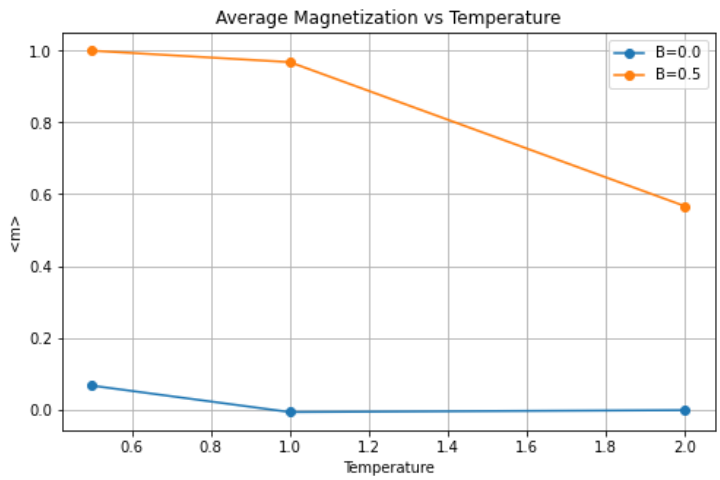

Figure 4: Average magnetization as a function of temperature for $B = 0$ and $B \neq 0$.

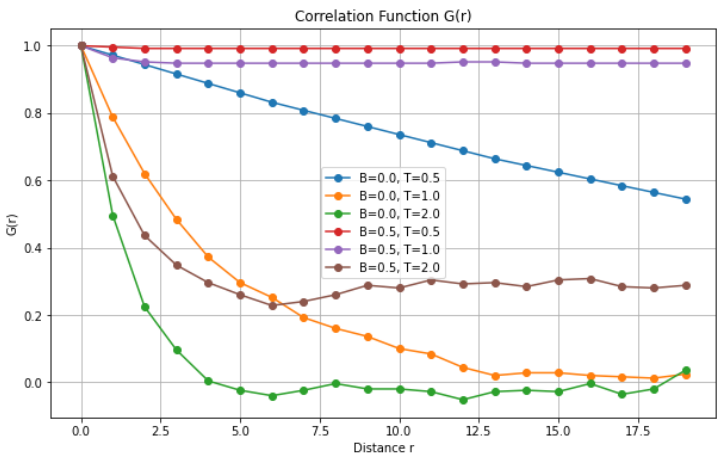

Figure 5: Correlation function $G(r)$ showing exponential decay and plateau behavior.

4 First-Principles MC Simulation of the 2D Ising Model

The two-dimensional Ising model provides the simplest lattice system exhibiting a genuine phase transition. Its Hamiltonian is:

$$E[s_i] = -J \sum_{\langle ij \rangle} s_i s_j - B \sum_i s_i. \tag{135}$$

In contrast to the one-dimensional case, the 2D Ising model possesses a non-trivial critical temperature:

$$T_c = \frac{2J}{\ln(1+\sqrt{2})}. \tag{136}$$

The system exhibits two distinct phases:

- **Ordered phase** ($T < T_c$): spontaneous magnetization,

$$\langle |m| \rangle \neq 0, \tag{137}$$

  indicating broken $\mathbb{Z}_2$ symmetry.

- **Disordered phase** ($T > T_c$): no long-range order,

$$\langle m \rangle = 0. \tag{138}$$

- **Critical point** ($T = T_c$): scale invariance and diverging correlation length,

$$\xi \to \infty. \tag{139}$$

The existence of a finite-temperature phase transition originates from the competition between the energy and entropy of domain walls, which scale with the same power of system size in two dimensions.

4.1 Monte Carlo Sampling and Observables

Thermal expectation values are computed via importance sampling:

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \sum_{s_i} \mathcal{O}[s_i] e^{-\beta E[s_i]}. \tag{140}$$

In practice, we generate configurations using a Markov chain and approximate:

$$\langle \mathcal{O} \rangle \approx \frac{1}{M} \sum_{t=1}^{M} \mathcal{O}(t). \tag{141}$$

The main idea is analogous to that of the one-dimensional case.

The primary observables are:

- Magnetization:

$$m = \frac{1}{N} \sum_i s_i, \tag{142}$$

- Absolute magnetization:

$$\langle |m| \rangle, \tag{143}$$

  which serves as the order parameter in finite systems,

- Susceptibility:

$$\chi = \frac{1}{T} N \left(\langle m^2 \rangle - \langle m \rangle^2 \right). \tag{144}$$

4.2 Wolff Cluster Algorithm

Near the critical point, local update algorithms such as Metropolis suffer from critical slowing down due to the emergence of large correlated domains. The Wolff algorithm overcomes this by updating entire clusters of aligned spins.

The algorithm proceeds as follows:

1. Select a random lattice site as the seed of a cluster,
2. Grow a cluster by adding nearest neighbors with the same spin orientation with probability:

$$P_{\text{add}} = 1 - e^{-2\beta J}, \tag{145}$$

3. Repeat the growth process until no new sites are added,
4. Flip all spins in the cluster simultaneously.

This construction satisfies detailed balance and efficiently samples configurations distributed according to the Boltzmann weight.

The Wolff algorithm provides several key advantages:

- **Reduction of critical slowing down:** cluster updates match the physical correlation length scale,
- **Efficient decorrelation:** large domains are flipped in a single update,
- **Improved sampling near T_c:** critical fluctuations are captured more accurately.

In contrast, the Metropolis algorithm updates spins locally and becomes inefficient when long-range correlations develop.

4.3 Numerical Signatures of the Phase Transition

The phase transition is identified through several numerical observables:

- **Magnetization:** $\langle |m| \rangle$ decreases sharply near T_c,
- **Susceptibility:** jumps discontinuously near T_c,
- **Spin configurations:** reveal domain formation below T_c and scale-invariant structures at criticality.

Although the non-analytic behavior occurs only in the thermodynamic limit $N \to \infty$, we can still characterize the properties of the phase transition near the critical point.

4.4 Numerical Results

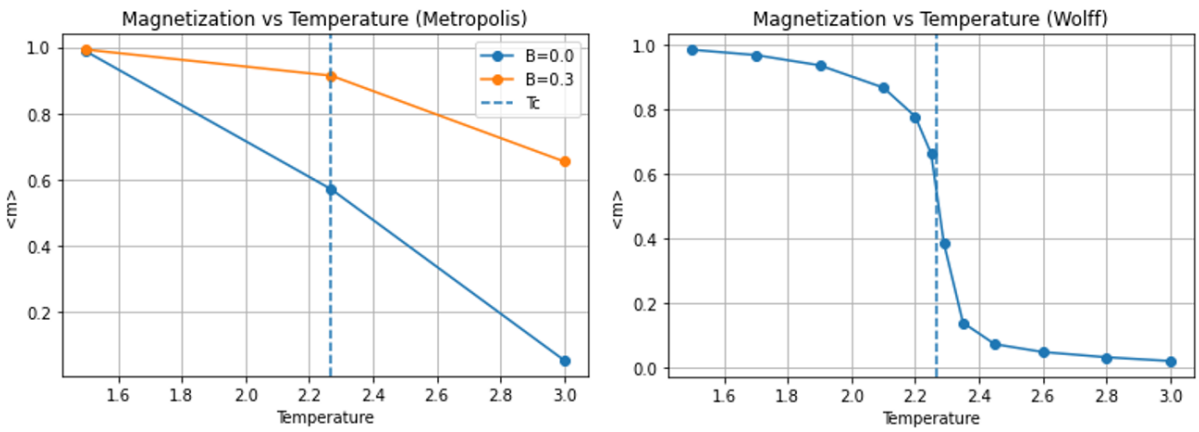

Figure 6: Magnetization as a function of temperature for both Metropolis and Wolff algorithms.

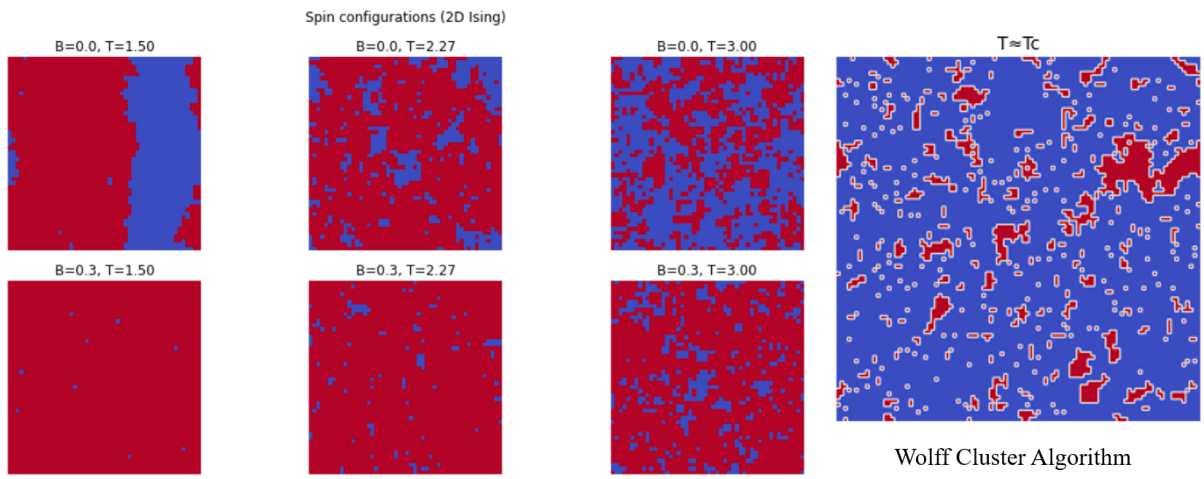

Figure 7: Spin configurations generated by the Metropolis algorithm (six subplots) and the Wolff algorithm at the critical temperature.

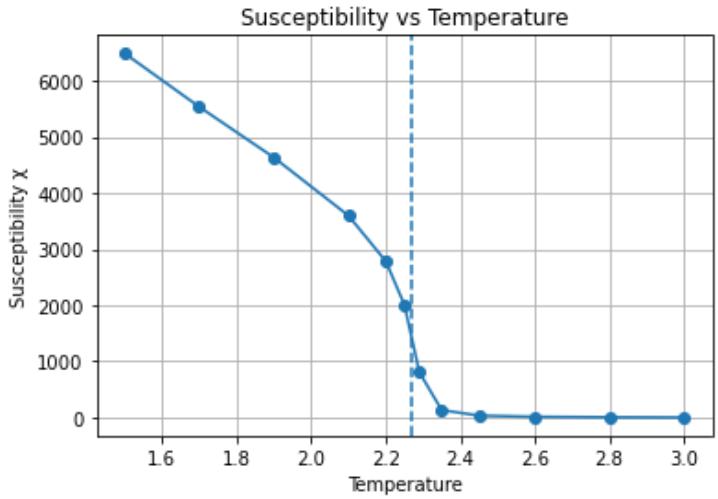

Figure 8: Susceptibility as a function of temperature obtained using the Wolff algorithm, showing a discontinuous jump at T_c.

The two-dimensional Ising model provides a concrete realization of phase transitions, spontaneous symmetry breaking, and critical phenomena. The Wolff algorithm enables efficient numerical exploration of these phenomena by directly incorporating the structure of correlated fluctuations into the update dynamics. This establishes a clear bridge between microscopic statistical models and the field-theoretic description of criticality.

Some of the derivations concerning the Ising model and Landau-Ginzburg theory are based on David Tong's lectures on statistical field theory. The code for the Ising model simulation is available on Github: `https://github.com/ChitsinYin/ManyBody-FieldTheory/blob/main/simulations/IsingMC_1D_2D.ipynb`.